



Board-Review Quality • Comprehensive Academic Reference

- Overview: The Skin as a Clinical Window
- 1 Chapters • Premium Tables • Key Teaching Points
- OSCE Pearls • MCQ Traps • High-Yield Facts

2025 Edition

TABLE OF CONTENTS

Chapter 1 Overview: The Skin as a Clinical Window

Overview: The Skin as a Clinical Window

The skin is far more than a passive covering — it is the body's largest organ and serves as a dynamic interface between the internal and external environments. It provides an essential living biological barrier, regulates body temperature through vasodilation and vasoconstriction, and prevents transepidermal water loss. Critically, the skin can serve as the first visible herald of underlying systemic disease; changes on the skin surface may be the earliest clue to conditions ranging from autoimmune disorders to internal malignancies. A thorough dermatologic examination should always include close inspection of the entire skin surface, nails, and mucous membranes, as pathology in any of these areas can provide diagnostic information. Furthermore, skin diseases carry a significant psychosocial burden — patients with visible or symptomatic skin conditions frequently experience anxiety, depression, and social isolation. The validated **Dermatology Life Quality Index (DLQI)** is the standard tool used to quantify the impact of skin disease on a patient's daily life, relationships, and emotional well-being.

KEY TEACHING POINTS

The skin is the largest organ of the body and serves as a barrier, thermoregulator, and water-loss preventer.

Skin changes may be the first sign of systemic disease — always examine the whole skin, nails, and mucous membranes.

The Dermatology Life Quality Index (DLQI) is a validated measure of how much skin disease affects patients' lives.

Psychiatric health is frequently impacted by dermatologic conditions; screen for depression and anxiety in chronic skin disease patients.

Skin Anatomy: The Three Layers

Understanding skin anatomy is fundamental to dermatologic diagnosis, as the layer involved determines the type of lesion, the clinical appearance, and the potential for scarring.

1. Epidermis

The epidermis is the outermost layer of the skin and is composed predominantly of **keratinocytes** — squamous epithelial cells that undergo progressive differentiation from the basal layer to the surface. The epidermis is **avascular**, receiving its nutrient supply by diffusion from the underlying dermis. It is rich in **keratin**, the structural protein that provides mechanical strength and waterproofing. The epidermis also contains **melanocytes** (pigment-producing cells), **Langerhans cells** (antigen-presenting immune cells), and **Merkel cells** (touch receptors).

2. Dermis

The dermis lies beneath the epidermis and is a connective tissue layer that provides structural support and nourishment. It contains collagen and elastin fibers, blood vessels, lymphatic channels, nerve endings, hair follicles, sebaceous glands, eccrine (sweat) glands, and apocrine glands. The dermis is divided into the superficial **papillary dermis** and the deeper **reticular dermis**. Pathology involving the dermis — such as nodules or deep ulcers — typically results in scarring because the regenerative capacity of dermal connective tissue is limited.

3. Subcutaneous Fat (Hypodermis Subcutis)

The subcutaneous fat layer, also known as the hypodermis or subcutis, lies beneath the dermis and is composed primarily of adipose tissue. It serves as an energy reservoir, provides thermal insulation, and acts as a mechanical cushion. Lesions that extend into this layer — such as deep nodules or panniculitis — can be particularly painful and may leave significant scarring upon resolution.

KEY TEACHING POINTS

The epidermis is avascular and rich in keratin; pathology confined to the epidermis generally heals without scarring.

The dermis contains blood vessels, nerves, hair follicles, and sweat glands; dermal injury typically scars.

The subcutaneous fat (hypodermis) provides insulation, cushioning, and energy storage.

Knowing which layer is involved helps predict whether a lesion will scar and guides biopsy depth decisions.

Skin Lesions: Primary and Secondary

A **skin lesion** refers to any area of skin that has different characteristics from the surrounding skin — whether in color, texture, elevation, or consistency. Skin lesions are divided into two main types: **primary** and **secondary**.

Primary skin lesions originate on previously healthy skin and are directly associated with a specific disease process. They represent the earliest visible manifestation of the underlying pathology.

Secondary skin lesions develop from the evolution of a primary skin lesion, either due to traumatic manipulation (such as scratching or rubbing), treatment effects, or natural disease progression. Recognizing whether a lesion is primary or secondary is essential for accurate diagnosis.

KEY TEACHING POINTS

Primary lesions arise on previously healthy skin and reflect the underlying disease directly.

Secondary lesions result from modification of primary lesions by scratching, infection, healing, or treatment.

Many patients present with mixed primary and secondary lesions — look for the "fresh" primary lesion to make the correct diagnosis.

Primary Skin Lesions: Morphology and Clinical Correlates

Macules and Patches

Macules and patches are **flat lesions** — they are not raised or palpable. They represent a change in skin color without any elevation above the surface of the surrounding skin. The color change may be erythematous (red), hypopigmented (lighter), or hyperpigmented (darker).

- **Macule:** < 1 cm in diameter. Example: Freckle (ephelis).
- **Patch:** > 1 cm in diameter. Example: Vitiligo, port-wine stain, café-au-lait macule (when >1 cm).

MCQ TRAPS & PITFALLS

A macule is flat and non-palpable — if you can feel it, it is NOT a macule.

Vitiligo presents as depigmented **patches** (not macules) because individual lesions are typically >1 cm.

Freckles are small **macules**, not papules — they are completely flat.

Macules vs. Patches — Key Distinguishing Features

Feature	Macule	Patch
Size	< 1 cm	> 1 cm
Elevation	Flat, non-palpable	Flat, non-palpable
Color change	Yes (erythematous, hypo-, or hyperpigmented)	Yes (erythematous, hypo-, or hyperpigmented)
Example	Freckle, petechia	Vitiligo, port-wine stain

Papules and Plaques

Papules and plaques are **raised, palpable solid lesions**. They are distinguished from macules and patches by the fact that they can be felt on palpation — there is a measurable elevation above the skin surface.

- **Papule:** < 1 cm in diameter, raised and palpable. Example: Acne papule, lichen planus papule, insect bite reaction.
- **Plaque:** > 1 cm in diameter, raised and palpable, often formed by the confluence (coalescence) of multiple papules. Example: Psoriasis plaque, mycosis fungoides plaque.

KEY TEACHING POINTS

Papules are small raised lesions < 1 cm; plaques are larger raised lesions > 1 cm.

Plaques often form when multiple papules merge together (confluence).

Psoriasis classically presents as well-demarcated, erythematous plaques with silvery-white scale.

Nodules and Tumors

- **Nodule:** A solid, palpable lesion > 1 cm in diameter that involves the dermis or subcutaneous tissue. Nodules are deeper and larger than papules. Example: Dermatofibroma, nodular melanoma, erythema nodosum.

- **Tumor:** A large solid mass arising from any tissue layer of the skin. The term does not imply malignancy — many tumors are benign. Example: Lipoma (benign fat tumor), dermatofibrosarcoma protuberans (malignant).

IMPORTANT

A nodule is deeper than a papule and involves the dermis or subcutis.

The word "tumor" in dermatology simply means a mass — it does NOT automatically mean cancer.

Nodular melanoma presents as a rapidly growing, often amelanotic (non-pigmented) nodule — a critical diagnosis not to miss.

Vesicles and Bullae

Vesicles and bullae are **fluid-filled blisters** — raised lesions containing clear serous fluid. They are distinguished primarily by size.

- **Vesicle:** < 0.5 cm in diameter, fluid-filled. Example: Herpes simplex (grouped vesicles on an erythematous base), herpes zoster, dyshidrotic eczema.
- **Bulla** (plural: bullae): > 0.5 cm in diameter, fluid-filled. Example: Bullous pemphigoid, pemphigus vulgaris, bullous impetigo, second-degree burn.

MCQ TRAPS & PITFALLS

Herpes simplex vesicles are classically described as "dewdrops on a rose petal" — small, grouped, on an erythematous base.

Bullous pemphigoid typically affects elderly patients and produces tense bullae.

Pemphigus vulgaris produces flaccid bullae that rupture easily — a key distinguishing feature.

Vesicles vs. Bullae — Clinical Comparison

Feature	Vesicle	Bulla
Size	< 0.5 cm	> 0.5 cm
Contents	Clear serous fluid	Clear serous fluid (may be hemorrhagic)
Wall tension	May be tense or flaccid	Tense (bullous pemphigoid) or flaccid (pemphigus vulgaris)
Example	Herpes simplex, contact dermatitis	Bullous pemphigoid, pemphigus vulgaris

Pustules

A **pustule** is a raised lesion containing **purulent (pus-containing) fluid** — as opposed to the clear serous fluid of vesicles and bullae. Pustules appear white or yellow against the surrounding skin.

- **Sterile pustule:** Pustule without bacterial infection. Example: Pustular psoriasis, candidal intertrigo.
- **Infectious pustule:** Pustule caused by bacterial infection. Example: Staphylococcal folliculitis, impetigo.

KEY TEACHING POINTS

Pustules contain pus (turbid/yellow fluid), distinguishing them from vesicles (clear fluid).

Not all pustules are infectious — pustular psoriasis produces sterile pustules.

Gram stain and culture of pustule contents can differentiate infectious from sterile causes.

Wheals (Urticaria)

A **wheel** is a transient, raised, erythematous plaque caused by **dermal edema**. Wheals are the hallmark of urticaria (hives) and are characteristically **pruritic** (itchy). Individual wheals typically last < 24 hours and resolve without scarring — if a single lesion persists > 24 hours, consider urticarial vasculitis.

- Example: Acute urticaria (allergic reaction), cholinergic urticaria, dermatographism.

IMPORTANT

Individual wheals in urticaria last < 24 hours — this is a key diagnostic criterion.

If a lesion persists > 24 hours, think urticarial vasculitis and biopsy.

Wheals resolve without scarring — scarring suggests vasculitis or another diagnosis.

Cysts

A **cyst** is a closed sac-like structure lined by epithelium, containing fluid, semi-solid material, or gas. In dermatology, the most common cyst is the **epidermoid cyst** (formerly called sebaceous cyst, though this is a misnomer).

- Example: Epidermoid cyst, pilar (trichilemmal) cyst, milium.

KEY TEACHING POINTS

Epidermoid cysts are lined by stratified squamous epithelium and filled with keratinous (cheesy) material.

The old term "sebaceous cyst" is a misnomer — true sebaceous cysts (steatocystoma) are rare.

Cysts may become inflamed or infected, presenting as a tender, erythematous nodule.

Secondary Skin Lesions: Morphology and Clinical Correlates

Scales (Desquamation)

Scale refers to visible flakes of the outermost layer of the epidermis (stratum corneum) that accumulate on the skin surface. Scaling results from abnormal keratinization or accelerated epidermal turnover.

- **Psoriasiform scale:** Thick, silvery-white, layered scale. Classic for psoriasis.
- **Pityriasis (fine bran-like scale):** Fine, white, powdery scale. Example: Pityriasis versicolor, pityriasis rosea.
- **Ichthyosiform scale:** Large, polygonal, fish-like scale. Example: Ichthyosis vulgaris.

KEY TEACHING POINTS

Scale = visible flakes of stratum corneum; indicates abnormal keratinization.

Psoriasis produces thick, silvery scale that can be scraped off (Auspitz sign — pinpoint bleeding when scale is removed).

Fine bran-like scale suggests pityriasis versicolor or pityriasis rosea.

Crusts (Scabs)

Crust is dried exudate — a combination of serum, blood, pus, and cellular debris that has dried on the skin surface. Crusts form when vesicles, bullae, or pustules rupture and the exudate dries.

- **Honey-colored crust:** Classic for impetigo (staphylococcal or streptococcal).
- **Hemorrhagic crust:** Dried blood; may be seen in excoriations or pemphigus vulgaris.

MCQ TRAPS & PITFALLS

Honey-colored crust is pathognomonic for **impetigo** — a high-yield association.

Crusts are secondary lesions — they form from ruptured primary lesions (vesicles, pustules).

Removing crusts may reveal an underlying erosion or ulcer.

Erosions

An **erosion** is a **superficial** loss of skin that involves only the epidermis. Because the dermis is not breached, erosions heal **without scarring**.

- Example: Ruptured vesicles in pemphigus vulgaris, superficial abrasions, areas where crusts have been removed.

Ulcers

An **ulcer** is a **deeper** loss of skin that extends into the dermis or subcutaneous tissue. Because the dermis is involved, ulcers heal **with scarring**.

- Example: Venous ulcers, arterial ulcers, diabetic foot ulcers, pyoderma gangrenosum, basal cell carcinoma (rodent ulcer).

KEY TEACHING POINTS

Erosions = epidermal loss only → heal without scarring.

Ulcers = dermal or deeper loss → heal with scarring.

The distinction between erosion and ulcer is critical for predicting scarring outcomes.

Erosions vs. Ulcers — Key Differences

Feature	Erosion	Ulcer
Depth	Epidermis only	Dermis or deeper
Scarring	No	Yes
Example	Ruptured pemphigus vesicle	Venous ulcer, pyoderma gangrenosum

Fissures

A **fissure** is a linear crack or split in the skin, typically occurring in areas of thickened or dry skin. Fissures are often painful and may bleed.

- Example: Angular cheilitis, fissured hand eczema, heel fissures in ichthyosis.

Atrophy

Atrophy refers to thinning of the skin, which may involve the epidermis, dermis, or both. Atrophic skin appears thin, wrinkled, and translucent, with visible underlying blood vessels.

- **Epidermal atrophy:** Skin appears thin and wrinkled (like cigarette paper). Example: Chronic corticosteroid use, aging.
- **Dermal atrophy:** Depression or indentation of the skin surface. Example: Striae (stretch marks), morphea.

IMPORTANT

Chronic topical corticosteroid use causes skin atrophy — counsel patients on appropriate duration and potency.

Striae represent dermal atrophy — they are permanent and appear as depressed, linear bands.

Scars and Keloids

A **scar** is the result of repair following dermal injury. Normal scars are flat and pale. A **keloid** is an abnormal scar that extends beyond the boundaries of the original wound and does not regress spontaneously.

- **Hypertrophic scar:** Raised but confined to the original wound boundary; may regress over time.
- **Keloid:** Extends beyond the original wound; does not regress; more common in darker skin types.

KEY TEACHING POINTS

Keloids extend beyond the original wound margin — this distinguishes them from hypertrophic scars.

Keloids are more common in individuals with darker skin pigmentation (Fitzpatrick types IV–VI).

Common sites for keloids: earlobes, chest, shoulders, upper back.

Lichenification

Lichenification is thickening of the skin with accentuation of normal skin markings, resulting from chronic rubbing or scratching. It represents the skin's response to persistent mechanical irritation.

- Example: Chronic eczema (atopic dermatitis), lichen simplex chronicus.

Excoriation

An **excoriation** is a linear erosion or ulcer caused by scratching. It is a secondary lesion that indicates pruritus (itching).

- Example: Scabies, atopic dermatitis, prurigo nodularis.

Descriptive Terminology in Dermatology

Dermatologists use a precise vocabulary to describe skin lesions. Mastering this terminology is essential for accurate clinical documentation and communication.

Lesion Configuration

The **configuration** (arrangement) of skin lesions provides important diagnostic clues.

- **Discrete:** Individual lesions that are separate and distinct from one another.
- **Grouped:** Lesions clustered together in a small area. Example: Herpes zoster (grouped vesicles in a dermatomal distribution).
- **Confluent:** Lesions that have merged together to form a larger continuous area. Example: Confluent papules forming a plaque in psoriasis.
- **Linear:** Lesions arranged in a line. Example: Contact dermatitis from poison ivy (linear streaks from plant contact), Koebner phenomenon in psoriasis.
- **Annular:** Ring-shaped lesions with central clearing. Example: Tinea corporis (ringworm), granuloma annulare, erythema migrans (Lyme disease).
- **Reticular:** Net-like or lacy pattern. Example: Livedo reticularis, erythema ab igne.
- **Targetoid (iris):** Lesions with three concentric zones. Example: Erythema multiforme (classic target lesions).

KEY TEACHING POINTS

Annular lesions with central clearing suggest tinea (perform KOH preparation to confirm).

Target lesions (3 concentric zones) are classic for erythema multiforme.

Linear lesions suggest external contact (poison ivy) or Koebner phenomenon.

Lesion Configuration Patterns and Associated Diagnoses

Configuration	Description	Associated Conditions
Discrete	Separate, individual lesions	Urticaria, drug eruption
Grouped	Clustered together	Herpes zoster, herpes simplex
Confluent	Merged into larger areas	Psoriasis, pityriasis rosea
Linear	Arranged in a line	Poison ivy dermatitis, Koebner phenomenon
Annular	Ring-shaped with central clearing	Tinea corporis, erythema migrans, granuloma annulare
Targetoid	Three concentric zones	Erythema multiforme
Reticular	Net-like, lacy pattern	Livedo reticularis, erythema ab igne

Lesion Distribution

The **distribution** of skin lesions on the body is another critical diagnostic clue.

- **Sun-exposed areas:** Face, neck, dorsal hands, V of chest. Example: Systemic lupus erythematosus (malar rash), dermatomyositis, porphyria cutanea tarda.
- **Flexural (intertriginous)** Skin folds — antecubital fossae, popliteal fossae, axillae, groin. Example: Atopic dermatitis, inverse psoriasis, candidal intertrigo.
- **Extensor surfaces:** Elbows, knees, shins. Example: Psoriasis, papular urticaria.
- **Dermatomal:** Following a single dermatome. Example: Herpes zoster.
- **Symmetrical:** Bilateral and symmetric. Example: Psoriasis, atopic dermatitis, drug eruptions.
- **Asymmetrical:** Unilateral or asymmetric. Example: Tinea (usually unilateral), contact dermatitis (depends on exposure).

MCQ TRAPS & PITFALLS

Herpes zoster follows a **dermatomal** distribution and does NOT cross the midline.

Psoriasis favors **extensor** surfaces (elbows, knees); atopic dermatitis favors **flexural** areas.

A symmetrical rash suggests a systemic cause (drug eruption, viral exanthem); an asymmetrical rash suggests a local cause (tinea, contact dermatitis).

Skin Color Changes

Erythema

Erythema is redness of the skin caused by increased blood flow (vasodilation) in the superficial dermal vasculature. It **blanches** on diascopy (pressure with a glass slide), distinguishing it from purpura.

- Example: Cellulitis, sunburn, drug eruption, viral exanthem.

Purpura

Purpura is red or purple discoloration of the skin caused by extravasation of red blood cells from blood vessels. It does **NOT blanch** on diascopy — this is the key distinguishing feature from erythema.

- **Petechiae:** < 3 mm, pinpoint purpuric lesions.
- **Ecchymoses:** > 1 cm, larger bruises.
- **Palpable purpura:** Raised purpuric lesions; suggests vasculitis.

KEY TEACHING POINTS

Blanching = erythema (increased blood flow); **Non-blanching** = purpura (extravasated blood).

Palpable purpura is the hallmark of **leukocytoclastic vasculitis**.

Non-blanching purpura in a child with fever and meningeal signs = meningococcal septicemia until proven otherwise.

Hyperpigmentation

Hyperpigmentation is darkening of the skin due to increased melanin deposition or other pigments.

- Example: Addison's disease (generalized hyperpigmentation), melasma, post-inflammatory hyperpigmentation, hemochromatosis (bronze skin).

Hypopigmentation and Depigmentation

- **Hypopigmentation:** Partial loss of skin color (lighter than surrounding skin). Example: Pityriasis versicolor, post-inflammatory hypopigmentation.
- **Depigmentation:** Complete loss of skin color (white). Example: Vitiligo.

IMPORTANT

Vitiligo causes **depigmentation** (complete loss of color), not just hypopigmentation.

Wood's lamp examination enhances the visibility of depigmented lesions in vitiligo.

Post-inflammatory hypopigmentation is common after resolution of inflammatory skin conditions, especially in darker skin types.

Diagnostic Techniques in Dermatology

Diascopy

Diascopy involves pressing a glass slide or transparent instrument against a skin lesion to determine whether it blanches. This simple bedside test distinguishes erythema (blanching) from purpura (non-blanching).

- **Blanching:** Erythema — increased blood flow in vessels.
- **Non-blanching:** Purpura — extravasated red blood cells.

Dermoscopy

Dermoscopy (dermatoscopy) is a non-invasive technique using a handheld magnifying device with polarized or non-polarized light to visualize subsurface skin structures not visible to the naked eye. It is particularly valuable in the evaluation of pigmented lesions and the early detection of melanoma.

- Example: Evaluating melanocytic nevi, distinguishing melanoma from benign pigmented lesions, diagnosing scabies (delta-wing jet sign).

KEY TEACHING POINTS

Dermoscopy improves diagnostic accuracy for melanoma by 20–30% compared to naked-eye examination alone.

The "ABCDE" criteria (Asymmetry, Border irregularity, Color variation, Diameter > 6mm, Evolution) are used to assess pigmented lesions.

Dermoscopy can reveal specific patterns (reticular, globular, starburst, blue-white structures) that aid diagnosis.

KOH Preparation

Potassium hydroxide (KOH) preparation is a rapid bedside test used to diagnose fungal skin infections. A skin scraping is placed on a slide with 10–20% KOH solution, which dissolves keratin and reveals fungal hyphae or spores under microscopy.

- Example: Tinea corporis, tinea pedis, tinea capitis, pityriasis versicolor.

Tzanck Smear

A **Tzanck smear** involves scraping the base of a fresh vesicle or bulla, staining the cells, and examining them under microscopy for multinucleated giant cells (which indicate herpesvirus infection) or acantholytic cells (which indicate pemphigus).

- Example: Herpes simplex, herpes zoster, pemphigus vulgaris.

MCQ TRAPS & PITFALLS

Tzanck smear shows **multinucleated giant cells** in herpes infections — but cannot distinguish HSV from VZV.

KOH preparation is the first-line test for suspected fungal skin infections — it is rapid, inexpensive, and highly effective.

A negative KOH does not completely rule out fungal infection — consider fungal culture if clinical suspicion remains high.

Skin Biopsy

Skin biopsy is the gold standard for definitive diagnosis of many dermatologic conditions. The two main types are:

- **Shave biopsy:** Superficial biopsy using a blade to sample the epidermis and superficial dermis. Suitable for superficial lesions, seborrheic keratoses, and some basal cell carcinomas.
- **Punch biopsy:** A circular blade (typically 3–6 mm) removes a core of skin including the epidermis, dermis, and sometimes subcutaneous fat. Suitable for deeper lesions, inflammatory dermatoses, and bullous diseases.
- **Excisional biopsy:** Complete removal of the lesion with a margin of normal skin. Preferred for suspected melanoma (to assess Breslow thickness).

KEY TEACHING POINTS

Punch biopsy is the most commonly performed skin biopsy in clinical practice.

For suspected melanoma, **excisional biopsy** is preferred to allow assessment of Breslow thickness.

Always include clinical information on the pathology request form — it significantly improves diagnostic accuracy.

Common Dermatologic Conditions: A Clinical Overview

Atopic Dermatitis (Eczema)

Atopic dermatitis is the most common chronic inflammatory skin disease, affecting up to 20% of children and 2–10% of adults. It is part of the **atopic triad** (eczema, allergic rhinitis, asthma) and is associated with a defective skin barrier, often due to mutations in the **filaggrin** gene.

Clinical features:

- Intense pruritus (itching) — often the most prominent symptom
- Erythematous, scaly, crusted patches and plaques
- Distribution varies by age: face and extensor surfaces in infants; flexural areas (antecubital and popliteal fossae) in children and adults
- Lichenification in chronic cases (from repeated scratching)
- Xerosis (dry skin) is nearly universal

Diagnostic criteria (Hanifin and Rajka): Pruritus + 3 or more of: history of flexural involvement, history of asthma/allergic rhinitis, history of dry skin, onset before age 2 years, visible flexural dermatitis.

KEY TEACHING POINTS

Atopic dermatitis is the most common chronic inflammatory skin disease.

The atopic triad includes eczema, allergic rhinitis, and asthma.

Filaggrin gene mutations impair the skin barrier, leading to increased transepidermal water loss and allergen penetration.

Emollients are the cornerstone of management — apply liberally and frequently.

Atopic Dermatitis — Age-Dependent Distribution

Age Group	Typical Distribution
Infants (0–2 years)	Face (especially cheeks), extensor surfaces, trunk
Children (2–12 years)	Flexural areas (antecubital fossae, popliteal fossae, wrists, ankles)
Adolescents/Adults	Flexural areas, hands, face (especially periorbital), neck

Psoriasis

Psoriasis is a chronic, immune-mediated inflammatory skin disease affecting approximately 2–3% of the global population. It is driven by T-cell activation and elevated levels of TNF- α and IL-17/IL-23.

Clinical features (plaque psoriasis — most common type):

- Well-demarcated, erythematous plaques with thick, silvery-white scale
- Favors extensor surfaces (elbows, knees), scalp, sacrum, and gluteal cleft

- **Auspitz sign:** Pinpoint bleeding when scale is removed
- **Koebner phenomenon:** New lesions develop at sites of skin trauma
- Nail involvement: pitting, onycholysis, oil-drop discoloration, subungual hyperkeratosis

Types of psoriasis:

- **Plaque psoriasis:** Most common (80–90% of cases)
- **Guttate psoriasis:** Small, drop-like lesions; often triggered by streptococcal pharyngitis in young adults
- **Pustular psoriasis:** Sterile pustules; may be localized or generalized (medical emergency)
- **Erythrodermic psoriasis:** >90% body surface area involvement; medical emergency
- **Inverse psoriasis:** Smooth, erythematous patches in skin folds (axillae, groin, inframammary)

IMPORTANT

Psoriasis is associated with several comorbidities: psoriatic arthritis (up to 30% of patients), cardiovascular disease, metabolic syndrome, depression, and inflammatory bowel disease.

Guttate psoriasis is often triggered by streptococcal infection — check ASO titers.

Erythrodermic psoriasis is a medical emergency requiring hospitalization.

Acne Vulgaris

Acne vulgaris is a chronic inflammatory disease of the pilosebaceous unit, affecting approximately 85% of adolescents. It is driven by four key factors: excess sebum production, follicular hyperkeratinization, colonization by *Cutibacterium acnes* (formerly *Propionibacterium acnes*), and inflammation.

Clinical features:

- **Comedones:** Open (blackheads) and closed (whiteheads) — the hallmark primary lesion
- **Inflammatory lesions:** Papules, pustules, nodules, cysts
- **Distribution:** Face, chest, upper back (areas with high sebaceous gland density)
- **Scarring:** Ice-pick scars, rolling scars, hypertrophic scars

Treatment approach:

- **Mild:** Topical retinoid (tretinoin, adapalene) + topical benzoyl peroxide
- **Moderate:** Add topical antibiotic (clindamycin) or oral antibiotic (doxycycline 100mg PO daily)
- **Severe (nodulocystic):** Oral isotretinoin 0.5–1 mg/kg/day for 15–20 weeks
- **Hormonal therapy** (combined oral contraceptive pill, spironolactone) for females with hormonal acne

KEY TEACHING POINTS

Comedones are the hallmark of acne — their presence distinguishes acne from other papulopustular conditions.

Isotretinoin is the most effective treatment for severe acne but is teratogenic — mandatory pregnancy prevention program (iPLEDGE in the US).

Do NOT prescribe tetracycline antibiotics to pregnant women or children < 8 years (causes tooth discoloration).

Contact Dermatitis

Contact dermatitis is an inflammatory skin reaction caused by direct exposure to an external substance. It is divided into two types:

- **Irritant contact dermatitis:** Direct toxic effect of a substance on the skin; no prior sensitization required.
Example: Detergents, solvents, frequent hand washing.
- **Allergic contact dermatitis:** Type IV delayed hypersensitivity reaction; requires prior sensitization.
Example: Nickel allergy, poison ivy (urushiol), fragrance, preservatives.

Clinical features:

- Erythema, edema, vesicles, and scaling in the area of contact
- **Linear streaks** suggest plant contact (poison ivy/oak)
- Distribution provides clues: hands (occupational), ears/neck (nickel in jewelry), feet (shoe materials)

MCQ TRAPS & PITFALLS

Poison ivy causes **linear streaks** of vesicles — a classic presentation.

Nickel allergy is the most common cause of allergic contact dermatitis worldwide.

Patch testing is the gold standard for identifying the causative allergen in allergic contact dermatitis.

Tinea (Dermatophytosis)

Tinea refers to superficial fungal infections caused by dermatophytes (Trichophyton, Microsporum, Epidermophyton species). The name is modified by the body site involved.

- **Tinea corporis:** Body — annular, scaly plaques with central clearing and an active, raised, scaly border
- **Tinea pedis:** Feet — interdigital scaling, maceration, or moccasin-type diffuse scaling
- **Tinea capitis:** Scalp — scaly patches with hair breakage (black dot pattern); most common in children
- **Tinea cruris:** Groin — erythematous, scaly plaques with central clearing; spares the scrotum (unlike candidal intertrigo)
- **Tinea unguium (onychomycosis):** Nails — thickened, discolored, brittle nails

KEY TEACHING POINTS

Tinea corporis is annular with central clearing — "ringworm" (though it is a fungal infection, not a worm).

KOH preparation confirms the diagnosis — look for branching hyphae.

Tinea capitis requires **systemic** antifungal therapy (griseofulvin or terbinafine) — topical therapy alone is insufficient.

Tinea cruris spares the scrotum; candidal intertrigo involves the scrotum — a useful distinguishing feature.

Tinea Infections by Body Site

Site	Name	Key Features
Body	Tinea corporis	Annular plaque with central clearing, raised scaly border

Site	Name	Key Features
Feet	Tinea pedis	Interdigital scaling, maceration, or moccasin pattern
Scalp	Tinea capitis	Scaly patches, hair breakage; requires systemic therapy
Groin	Tinea cruris	Erythematous plaque, spares scrotum
Nails	Tinea unguium	Thickened, discolored, brittle nails

Impetigo

Impetigo is the most common bacterial skin infection in children, caused by *Staphylococcus aureus* or *Streptococcus pyogenes* (Group A Streptococcus).

Clinical features:

- **Non-bullous impetigo (70% of cases):** Erythematous macule → vesicle/pustule → ruptures → **honey-colored crust**
- **Bullous impetigo:** Flaccid bullae that rupture, leaving a thin brown crust; caused by *S. aureus* exfoliative toxins
- Most common on the face (perioral, perinasal) and extremities
- Highly contagious — spreads by direct contact

Treatment:

- Localized: Topical mupirocin 2% ointment TID x 5 days
- Extensive or systemic symptoms: Oral antibiotics (cephalexin 500mg PO BID x 7 days, or amoxicillin-clavulanate)

IMPORTANT

Honey-colored crust is the hallmark of impetigo — a high-yield association.

Post-streptococcal glomerulonephritis is a potential complication of impetigo caused by nephritogenic strains of Group A Strep.

Impetigo does NOT cause rheumatic fever (unlike streptococcal pharyngitis).

Herpes Simplex Virus (HSV)

Herpes simplex virus (HSV-1 and HSV-2) causes recurrent vesicular eruptions. HSV-1 typically causes orolabial herpes; HSV-2 typically causes genital herpes, though either type can infect either site.

Clinical features:

- **Primary infection:** Painful grouped vesicles on an erythematous base → ulcerate → crust over; may be associated with fever, malaise, regional lymphadenopathy
- **Recurrent infection:** Milder, shorter duration; often preceded by a prodrome of tingling, burning, or itching

- **Herpetic whitlow:** HSV infection of the finger (healthcare workers, children)
- **Eczema herpeticum:** Disseminated HSV infection in patients with pre-existing skin disease (especially atopic dermatitis) — a medical emergency

Treatment:

- Acyclovir 400mg PO TID x 7–10 days (or valacyclovir 1g PO BID x 7–10 days)
- Suppressive therapy for frequent recurrences: acyclovir 400mg PO BID

KEY TEACHING POINTS

HSV produces grouped vesicles on an erythematous base — "dewdrops on a rose petal."

Tzanck smear shows multinucleated giant cells but cannot distinguish HSV-1 from HSV-2.

Eczema herpeticum is a medical emergency — treat with IV acyclovir immediately.

Herpes Zoster (Shingles)

Herpes zoster results from reactivation of latent varicella-zoster virus (VZV) in dorsal root or cranial nerve ganglia.

Clinical features:

- Painful, unilateral, dermatomal eruption of grouped vesicles on an erythematous base
- Does **NOT cross the midline**
- Thoracic dermatomes are most commonly affected
- **Prodromal pain** may precede the rash by several days
- **Ramsay Hunt syndrome:** Herpes zoster of the facial nerve (CN VII) — vesicles on the ear, facial paralysis, hearing loss
- **Postherpetic neuralgia:** Persistent pain > 90 days after rash onset; most common complication, especially in elderly patients

Treatment:

- Antiviral therapy within 72 hours of rash onset: valacyclovir 1g PO TID x 7 days (preferred) or acyclovir 800mg PO 5x/day x 7 days
- Pain management: gabapentin, pregabalin, tricyclic antidepressants for postherpetic neuralgia
- Zoster vaccine (Shingrix) recommended for adults ≥ 50 years

IMPORTANT

Herpes zoster does NOT cross the midline — if the rash crosses the midline, reconsider the diagnosis.

Antivirals are most effective when started within 72 hours of rash onset.

Disseminated zoster (involving ≥ 3 dermatomes or visceral organs) in immunocompromised patients requires IV acyclovir.

Pemphigus Vulgaris

Pemphigus vulgaris is an autoimmune blistering disease caused by IgG autoantibodies against **desmoglein 3** (and sometimes desmoglein 1), which are components of desmosomes that hold keratinocytes together.

Clinical features:

- **Flaccid bullae** that rupture easily, leaving painful erosions
- **Nikolsky sign:** Lateral pressure on perilesional skin causes epidermal separation (positive)
- **Mucosal involvement:** Oral erosions are present in > 50% of patients and may be the initial presentation
- **Asboe-Hansen sign:** Pressure on an intact bulla causes lateral spread of the blister

Diagnosis:

- Skin biopsy: **suprabasal acantholysis** (loss of cell-cell adhesion above the basal layer)
- Direct immunofluorescence: IgG and C3 in a **chicken-wire** pattern around keratinocytes
- Indirect immunofluorescence: circulating anti-desmoglein antibodies

KEY TEACHING POINTS

Pemphigus vulgaris produces **flaccid** bullae (vs. bullous pemphigoid which produces **tense** bullae).

Nikolsky sign is positive in pemphigus vulgaris — lateral pressure causes epidermal separation.

Mucosal involvement is common and may precede skin lesions.

Before the era of corticosteroids, pemphigus vulgaris had a mortality rate of ~75%.

Bullous Pemphigoid

Bullous pemphigoid is the most common autoimmune blistering disease, typically affecting elderly patients (> 70 years). It is caused by IgG autoantibodies against **BP180** and **BP230**, components of hemidesmosomes at the dermal-epidermal junction.

Clinical features:

- **Tense bullae** on an erythematous or urticarial base
- Nikolsky sign is **negative** (distinguishing it from pemphigus vulgaris)
- Pruritus may be severe and may precede the bullous eruption by weeks to months
- Mucosal involvement is uncommon (< 20%)
- Distribution: trunk, inner thighs, flexural areas

Diagnosis:

- Skin biopsy: subepidermal blister with eosinophils
- Direct immunofluorescence: linear IgG and C3 along the **basement membrane zone**
- Indirect immunofluorescence: circulating anti-BP180/BP230 antibodies

Pemphigus Vulgaris vs. Bullous Pemphigoid — Key Comparisons

Feature	Pemphigus Vulgaris	Bullous Pemphigoid
Age	40–60 years	> 70 years
Blister type	Flaccid, easily ruptured	Tense, intact
Nikolsky sign	Positive	Negative

Feature	Pemphigus Vulgaris	Bullous Pemphigoid
Mucosal involvement	Common (> 50%)	Uncommon (< 20%)
Acantholysis	Suprabasal	None (subepidermal split)
DIF pattern	Chicken-wire (intercellular)	Linear BMZ
Target antigen	Desmoglein 3 (and 1)	BP180, BP230

Skin Cancer: Recognition and Management

Basal Cell Carcinoma (BCC)

Basal cell carcinoma is the most common cancer in humans, accounting for approximately 80% of all non-melanoma skin cancers. It arises from basal keratinocytes of the epidermis and is strongly associated with chronic UV exposure.

Clinical features:

- **Nodular BCC (most common subtype):** Pearly, translucent papule or nodule with telangiectasias, rolled borders, and central ulceration ("rodent ulcer")
- **Superficial BCC:** Erythematous, scaly patch or plaque; may mimic eczema or psoriasis
- **Morpheaform BCC:** Scar-like, indurated plaque; aggressive with poorly defined margins
- Metastasis is extremely rare (< 0.1%)

Risk factors: Fair skin (Fitzpatrick types I–II), chronic UV exposure, immunosuppression, Gorlin syndrome (nevoid basal cell carcinoma syndrome)

Treatment:

- Excision with 4–5 mm margins (standard for most BCCs)
- Mohs micrographic surgery (for high-risk locations: face, ears, nose; or recurrent tumors)
- Curettage and electrodesiccation, cryotherapy, topical imiquimod, or photodynamic therapy for low-risk superficial BCCs

KEY TEACHING POINTS

BCC is the most common cancer in humans — strongly associated with UV exposure.

The classic description is a pearly papule with telangiectasias and rolled borders.

BCC rarely metastasizes but can cause significant local destruction if untreated.

Mohs surgery provides the highest cure rate (99% for primary BCC) while preserving maximum normal tissue.

Squamous Cell Carcinoma (SCC)

Squamous cell carcinoma is the second most common skin cancer, arising from keratinocytes of the epidermis. Unlike BCC, SCC has a meaningful risk of metastasis (2–6%, higher in immunosuppressed

patients and high-risk locations).

Clinical features:

- Erythematous, hyperkeratotic (scaly/crusted) papule, plaque, or nodule
- May arise from a **pre-existing actinic keratosis** (pre-malignant lesion)
- Common on sun-exposed areas: face (especially ears and lower lip), dorsal hands, forearms
- **Marjolin ulcer:** SCC arising in a chronic wound or scar

Risk factors: Chronic UV exposure, fair skin, immunosuppression (especially organ transplant recipients), HPV infection, chronic wounds/scars, arsenic exposure

Treatment:

- Excision with 4–6 mm margins
- Mohs micrographic surgery for high-risk SCCs
- Radiation therapy for non-surgical candidates
- Sentinel lymph node biopsy for high-risk SCCs (deep, poorly differentiated, perineural invasion)

IMPORTANT

SCC can metastasize — unlike BCC — so early detection and treatment are critical.

Actinic keratoses are pre-malignant lesions that may progress to SCC — treat them.

Immunosuppressed patients (organ transplant recipients) have a 65–250x increased risk of SCC.

Melanoma

Melanoma is the most dangerous form of skin cancer, arising from melanocytes. Although it accounts for only ~1% of skin cancers, it is responsible for the majority of skin cancer deaths. Early detection is critical — the 5-year survival rate for localized melanoma is > 99%, but drops to < 30% for distant metastatic disease.

ABCDE criteria for detection:

- **Asymmetry:** One half of the lesion does not match the other
- **Border irregularity:** Ragged, notched, or blurred edges
- **Color variation:** Multiple colors (brown, black, tan, red, white, blue)
- **Diameter:** > 6 mm (pencil eraser size), though melanomas can be smaller
- **Evolving:** Changing in size, shape, color, or feel

Subtypes of melanoma:

- **Superficial spreading melanoma:** Most common (70%); grows horizontally (radial growth phase) before invading vertically
- **Nodular melanoma:** Second most common (15–30%); grows vertically from the outset; often amelanotic (non-pigmented); worse prognosis
- **Lentigo maligna melanoma:** Arises from lentigo maligna (melanoma in situ) on chronically sun-damaged skin; most common in elderly patients
- **Acral lentiginous melanoma:** Most common melanoma in darker skin types; occurs on palms, soles, and under nails (subungual melanoma)

KEY TEACHING POINTS

Melanoma is the most dangerous skin cancer — responsible for most skin cancer deaths.

The ABCDE criteria are used for screening — but melanomas can be < 6 mm or amelanotic.

Nodular melanoma is often amelanotic and may be misdiagnosed as a benign lesion — maintain a high index of suspicion.

Acral lentiginous melanoma is the most common melanoma in individuals with darker skin types — check palms, soles, and nail beds.

Melanoma Subtypes — Clinical Comparison

Subtype	Frequency	Key Features	Prognosis
Superficial spreading	70%	Radial growth phase; irregular borders; color variation	Better (if detected early)
Nodular	15–30%	Vertical growth from outset; often amelanotic	Worse (deeper at diagnosis)
Lentigo maligna	5–10%	Sun-damaged skin; elderly; large, irregularly pigmented macule	Good (if in situ)
Acral lentiginous	2–8%	Palms, soles, nail beds; most common in darker skin types	Variable

Skin Signs of Systemic Disease

The skin can provide critical clues to underlying systemic disease. Recognizing these associations is essential for early diagnosis and management.

Acanthosis Nigricans

Clinical features: Velvety, hyperpigmented thickening of the skin, typically in flexural areas (axillae, neck, groin).

Associations:

- **Benign (most common):** Insulin resistance, obesity, type 2 diabetes mellitus, metabolic syndrome
- **Malignant:** Gastric adenocarcinoma (rapid onset, extensive, may involve mucous membranes and palms)

KEY TEACHING POINTS

Acanthosis nigricans in an obese patient suggests insulin resistance.

New-onset, rapidly progressive acanthosis nigricans in a thin, elderly patient should raise suspicion for gastric carcinoma.

Check fasting glucose and HbA1c in patients with acanthosis nigricans.

Necrolytic Migratory Erythema

Clinical features: Erythematous, erosive, crusted plaques that migrate and coalesce, typically in perineural and intertriginous areas. Associated with glossitis and angular cheilitis.

Association: Glucagonoma (pancreatic neuroendocrine tumor) — part of the "4D syndrome": Dermatitis (NME), Diabetes, Deep vein thrombosis, Depression.

Dermatomyositis Skin Findings

Clinical features:

- **Heliotrope rash:** Violaceous/purplish discoloration of the eyelids with periorbital edema
- **Gottron papules:** Violaceous, flat-topped papules over the knuckles (MCP, PIP, DIP joints)
- **Shawl sign:** Erythema over the upper back and shoulders (shawl distribution)
- **Mechanic's hands:** Cracked, fissured skin on the palms and fingers

Association: Dermatomyositis (inflammatory myopathy); 15–30% of adult patients have an underlying malignancy (especially ovarian, lung, GI cancers).

IMPORTANT

Dermatomyositis skin findings may precede muscle symptoms by months.

All adults with newly diagnosed dermatomyositis should undergo age-appropriate cancer screening.

Gottron papules are pathognomonic for dermatomyositis — a high-yield association.

Erythema Nodosum

Clinical features: Tender, erythematous, subcutaneous nodules, typically on the anterior shins. The nodules are deep (involving subcutaneous fat) and do not ulcerate.

Associations:

- Infections: Streptococcal pharyngitis, tuberculosis, histoplasmosis, coccidioidomycosis
- Inflammatory: Inflammatory bowel disease (Crohn's disease > ulcerative colitis), sarcoidosis, Behçet's disease
- Drugs: Sulfonamides, oral contraceptive pills, penicillin
- Idiopathic (up to 55% of cases)

KEY TEACHING POINTS

Erythema nodosum is the most common form of panniculitis.

The nodules are tender and located on the shins — they do NOT ulcerate.

Always investigate for underlying causes: throat CXR, ASO titer, stool studies, TB testing.

Pyoderma Gangrenosum

Clinical features: Rapidly progressive, painful ulcer with violaceous, undermined borders. Often begins as a pustule or nodule that breaks down into an ulcer. **Pathergy** (new lesions at sites of minor trauma) is characteristic.

Associations: Inflammatory bowel disease (ulcerative colitis > Crohn's disease), rheumatoid arthritis, hematological malignancy (myelodysplastic syndrome, acute myeloid leukemia), monoclonal gammopathy.

IMPORTANT

Pyoderma gangrenosum is a diagnosis of exclusion — rule out infection, vasculitis, and malignancy first.

Pathergy is characteristic — do NOT debride or surgically intervene (worsens the condition).

Treatment is immunosuppressive (corticosteroids, cyclosporine, TNF- α inhibitors).

Skin Signs and Their Systemic Associations

Skin Finding	Key Features	Systemic Associations
Acanthosis nigricans	Velvety hyperpigmentation in flexures	Insulin resistance, gastric carcinoma
Necrolytic migratory erythema	Migratory erythematous erosive plaques	Glucagonoma
Heliotrope rash	Violaceous eyelid discoloration	Dermatomyositis (screen for malignancy)
Gotttron papules	Violaceous papules over knuckles	Dermatomyositis
Erythema nodosum	Tender subcutaneous nodules on shins	IBD, sarcoidosis, TB, strep, Behçet's
Pyoderma gangrenosum	Painful ulcer with undermined violaceous borders	IBD, RA, hematologic malignancy

Pharmacology in Dermatology

Topical Corticosteroids

Topical corticosteroids are the most commonly prescribed medications in dermatology. They are classified by potency (Group I = superpotent to Group VII = least potent).

Potency classification:

- **Superpotent (Group I):** Clobetasol propionate 0.05% — for thick, recalcitrant plaques (psoriasis on palms/soles)
- **High potency (Group II–III):** Betamethasone dipropionate 0.05%, triamcinolone acetonide 0.1% — for moderate-severe dermatitis on body
- **Medium potency (Group IV–V):** Hydrocortisone butyrate 0.1%, desonide 0.05% — for mild-moderate dermatitis
- **Low potency (Group VI–VII):** Hydrocortisone 1% (OTC) — for face, skin folds, children

Side effects of chronic use: Skin atrophy, striae, telangiectasias, acneiform eruption, hypopigmentation, rebound flare, adrenal suppression (with prolonged use of potent steroids over large body surface area).

KEY TEACHING POINTS

Use low-potency steroids on the face and skin folds; reserve superpotent steroids for thick plaques on palms and soles.

The fingertip unit (FTU) is a practical measure: 1 FTU = 0.5g, sufficient to cover 2 adult handprints (~2% BSA).

Chronic use of potent topical steroids causes skin atrophy — counsel patients on appropriate duration.

Topical Calcineurin Inhibitors

Tacrolimus (Protopic) and **pimecrolimus** (Elidel) are non-steroidal anti-inflammatory agents that inhibit calcineurin, reducing T-cell activation. They are used as steroid-sparing agents, particularly for facial and intertriginous dermatitis.

- Advantages: No skin atrophy — safe for long-term use on face and skin folds
- Disadvantages: Burning/stinging on application; black box warning (theoretical malignancy risk based on systemic use, though not confirmed with topical use)

Systemic Agents

Systemic Dermatologic Medications

Drug	Indication	Key Side Effects
Oral corticosteroids	Severe autoimmune blistering diseases, erythroderma, severe drug reactions	Osteoporosis, diabetes, adrenal suppression, infection risk
Methotrexate	Psoriasis, dermatomyositis	Hepatotoxicity, myelosuppression, teratogenic; supplement with folic acid
Cyclosporine	Severe psoriasis, pyoderma gangrenosum	Nephrotoxicity, hypertension, infection risk
Azathioprine	Autoimmune blistering diseases, vasculitis	Myelosuppression, hepatotoxicity; check TPMT before starting
Isotretinoin	Severe nodulocystic acne	Teratogenic (iPLEDGE), hypertriglyceridemia, dry mucous membranes, mood changes
Dapsone	Dermatitis herpetiformis, leprosy	Hemolytic anemia (check G6PD), methemoglobinemia, dapsone syndrome
Hydroxychloroquine	Cutaneous lupus, dermatomyositis	Retinopathy (annual ophthalmologic screening required)

OSCE Pearls: Dermatology Examination

OSCE PEARLS

Always examine the **entire skin surface**, including the scalp, nails, and mucous membranes — pathology may be hidden.

Use the **ABCDE criteria** when assessing pigmented lesions for melanoma.

Describe lesions using precise terminology: macule, papule, plaque, nodule, vesicle, bulla, pustule, wheal.

Note the **distribution** (symmetrical vs. asymmetrical, sun-exposed vs. flexural) and **configuration** (annular, linear, grouped, dermatomal).

Perform **diascopy** to distinguish erythema (blanching) from purpura (non-blanching).

Check for **Nikolsky sign** in any patient with blisters or erosions.

Always ask about **medications** — drug eruptions are a common cause of skin rashes.

Ask about **travel history**, **occupational exposures**, and **contact history** (e.g., TB, scabies).

In a patient with a new rash, always check for **mucosal involvement** (oral, genital, conjunctival) — this can be a critical diagnostic clue.

Practice Questions

Q: A 45-year-old man presents with a 2-week history of a slowly enlarging, painless nodule on his nose. The lesion is a 1.5 cm pearly, flesh-colored papule with telangiectasias on its surface and a central ulcer. He has a history of chronic sun exposure. What is the most likely diagnosis?

- A. Squamous cell carcinoma
- B. Basal cell carcinoma
- C. Melanoma
- D. Seborrheic keratosis
- E. Actinic keratosis

Answer: B

Explanation: The classic description of basal cell carcinoma is a pearly, translucent papule or nodule with telangiectasias, rolled borders, and central ulceration ("rodent ulcer"). BCC is the most common skin cancer and is strongly associated with chronic UV exposure. SCC typically presents as a hyperkeratotic, crusted lesion rather than a pearly nodule. Melanoma would typically show color variation and irregular borders.

Q: A 72-year-old woman presents with a 3-month history of intensely pruritic urticarial plaques on her trunk and proximal extremities, followed by the development of tense bullae. Nikolsky sign is negative. Direct immunofluorescence shows linear IgG and C3 deposition along the basement membrane zone. What is the most likely diagnosis?

- A. Pemphigus vulgaris
- B. Bullous pemphigoid

- C. Dermatitis herpetiformis
- D. Linear IgA bullous dermatosis
- E. Epidermolysis bullosa acquisita

Answer: B

Explanation: Bullous pemphigoid is the most common autoimmune blistering disease, typically affecting elderly patients. It is characterized by tense bullae (not flaccid), negative Nikolsky sign, and linear IgG/C3 deposition along the basement membrane zone on direct immunofluorescence. Pemphigus vulgaris produces flaccid bullae with a positive Nikolsky sign and intercellular (chicken-wire) IgG pattern on DIF.

Q: A 19-year-old college student presents with a 5-day history of a painful, unilateral rash on his left chest. The rash consists of grouped vesicles on an erythematous base in a dermatomal distribution. He reports severe burning pain in the area. What is the most appropriate treatment?

- A. Oral acyclovir 200mg 5x/day x 5 days
- B. Oral valacyclovir 1g PO TID x 7 days
- C. Topical acyclovir 5% cream
- D. Oral fluconazole 200mg PO daily x 14 days
- E. Topical mupirocin 2% ointment TID x 5 days

Answer: B

Explanation: This presentation is classic for herpes zoster (shingles) — painful, unilateral, dermatomal grouped vesicles. Treatment is with oral antiviral therapy, ideally within 72 hours of rash onset. Valacyclovir 1g PO TID x 7 days is preferred over acyclovir due to better bioavailability and simpler dosing. Topical antivirals are insufficient for herpes zoster. Fluconazole is an antifungal, and mupirocin is a topical antibacterial — neither is appropriate.

Q: A 35-year-old woman presents with violaceous, flat-topped, pruritic papules on her wrists and ankles. She also has white, lacy streaks on the buccal mucosa. A skin biopsy shows a band-like lymphocytic infiltrate at the dermal-epidermal junction. What is the most likely diagnosis?

- A. Psoriasis
- B. Lichen planus
- C. Discoid lupus erythematosus
- D. Pityriasis rosea
- E. Secondary syphilis

Answer: B

Explanation: Lichen planus presents with violaceous (purple), flat-topped, pruritic papules, typically on the wrists, ankles, and lower back. The characteristic mucosal finding is white, lacy Wickham striae on the buccal mucosa. Histopathology shows a band-like lymphocytic infiltrate at the dermal-epidermal junction (interface dermatitis). The classic description is "5 P's": Purple, Polygonal, Pruritic, Planar (flat-topped), Papules.

Q: A 60-year-old man with a history of chronic sun exposure presents with a rapidly growing, non-pigmented nodule on his back. The lesion is 2 cm in diameter, firm, and dome-shaped. There is no color variation. Which subtype of melanoma is most likely?

- A. Superficial spreading melanoma
- B. Lentigo maligna melanoma
- C. Acral lentiginous melanoma
- D. Nodular melanoma
- E. Amelanotic melanoma is not a recognized subtype

Answer: D

Explanation: Nodular melanoma is the second most common subtype and is characterized by vertical growth from the outset (no radial growth phase). It often presents as a rapidly growing, firm, dome-shaped nodule that may be amelanotic (non-pigmented). This subtype has a worse prognosis because it is deeper at the time of diagnosis. While amelanotic melanoma is a recognized clinical variant, nodular melanoma is the most likely subtype given the clinical description.

Q: A 28-year-old woman presents with tender, erythematous nodules on both shins. She reports a sore throat 2 weeks ago. Chest X-ray is normal. ASO titer is elevated. What is the most likely diagnosis?

- A. Cellulitis
- B. Erythema nodosum
- C. Leukocytoclastic vasculitis
- D. Superficial thrombophlebitis
- E. Erythema multiforme

Answer: B

Explanation: Erythema nodosum presents as tender, erythematous, subcutaneous nodules on the anterior shins. It is the most common form of panniculitis. The most common identifiable cause in this age group is streptococcal pharyngitis, supported by the elevated ASO titer. The nodules are deep (involving subcutaneous fat) and do not ulcerate. Cellulitis is typically unilateral and associated with warmth and systemic signs of infection. Vasculitis presents with palpable purpura, not deep nodules.

Q: A 55-year-old man with a history of ulcerative colitis presents with a rapidly enlarging, painful ulcer on his left leg. The ulcer has violaceous, undermined borders. It began as a pustule 1 week ago. What is the most important next step in management?

- A. Surgical debridement of the ulcer
- B. Skin biopsy to confirm the diagnosis and rule out other causes
- C. Intravenous antibiotics for presumed cellulitis
- D. Compression therapy for presumed venous ulcer
- E. Topical corticosteroids

Answer: B

*Explanation: This presentation is classic for pyoderma gangrenosum — a rapidly progressive, painful ulcer with violaceous, undermined borders in a patient with inflammatory bowel disease. The most important next step is skin biopsy to confirm the diagnosis and rule out infection, vasculitis, and malignancy. **Surgical debridement is contraindicated** because of pathergy — the wound will worsen with surgical intervention. Treatment is immunosuppressive (corticosteroids, cyclosporine).*

Quick Reference: High-Yield Dermatology

High-Yield Dermatology Associations

Clinical Finding	Diagnosis
Honey-colored crust	Impetigo
Target lesions (3 zones)	Erythema multiforme
Silvery scale on extensor surfaces	Psoriasis
Grouped vesicles on erythematous base	Herpes simplex / zoster
Flaccid bullae, positive Nikolsky sign	Pemphigus vulgaris
Tense bullae, negative Nikolsky sign	Bullous pemphigoid
Pearly papule with telangiectasias	Basal cell carcinoma
Violaceous papules over knuckles	Dermatomyositis (Gottron papules)
Violaceous eyelid rash	Dermatomyro